# Ultrasonic Waveform P and S Arrival Picker

## Student installation and use guide

**Purpose.** This graphical tool displays a folder of ultrasonic waveforms, applies a zero-phase low-pass filter, and records manually selected P- and S-wave arrival times. The final output is a tab-separated text file containing one row per waveform.

**What you will produce.** A file named waveform_picks_summary.txt with the P and S picks in microseconds. The original waveform files are read but never modified.

## Workflow at a glance

1. Create and activate a dedicated Anaconda environment.
2. Install the required Python packages and launch wave_analyzer.py.
3. Select the folder that contains the waveform files.
4. Choose a filter cutoff, zoom if necessary, and pick the P and S onsets.
5. Save the summary file, then apply the experimental time calibration before calculating velocity.

## What the GUI does and does not do

- Displays one waveform at a time and processes files in alphabetical order.
- Applies a fourth-order Butterworth low-pass filter using zero-phase filtering.
- Stores one P pick and one S pick for each file.
- Does not calculate velocity, subtract apparatus delay, or export plots.
- Does not include a Previous button. Check each waveform before selecting Next Waveform.

# 1 Installation

These instructions assume that Anaconda is already installed. Use Anaconda Prompt on Windows or a Terminal with conda available on macOS and Linux.

## Create the environment

```
conda create -n waveform_picker python=3.11 -y
conda activate waveform_picker
```

## Install the required packages

```
conda install numpy pandas matplotlib scipy tk -y
```

The environment contains the numerical libraries, plotting tools, signal-processing functions, and the Tk graphical interface required by the program.

## Verify the installation

```
python -c "import tkinter, numpy, pandas, matplotlib, scipy; print('Installation
complete')"
```

If the message Installation complete appears without an error, the environment is ready.

## Launch the GUI

Place wave_analyzer.py in a convenient folder. Then move to that folder and run it. **Windows**

```
cd /d "C:\path\to\the\folder"
python wave_analyzer.py
```

**macOS or Linux**

```
cd "/path/to/the/folder"
python wave_analyzer.py
```

Activate the waveform_picker environment again whenever you open a new terminal before launching the program.

## Quick installation check

- A window titled Ultrasonic Waveform P and S Arrival Picker opens.
- The top row contains Select Folder, Filter Cutoff, and Apply Filter.
- The second row contains P-wave, S-wave, Save All Picks, and Next Waveform.

# 2 Prepare the waveform files

Put all waveforms that belong to one picking session in the same folder. The program reads every regular, non-hidden file in that folder, regardless of its filename extension. Subfolders are not searched.

## Required file structure

Each waveform must contain at least two numeric columns. The program interprets the columns in this order:

| Column | Content | Requirement |
|---|---|---|
| 0 | Amplitude | Numeric signal amplitude; arbitrary units are acceptable. |
| 1 | Time | Use seconds or microseconds consistently across the whole folder. |

A comma, tab, semicolon, or whitespace delimiter can normally be detected automatically. A header is not required; any non-numeric row is discarded. A simple comma-separated example is:

```
0.0021,0.0000000
0.0028,0.0000002
0.0034,0.0000004
```

## Time-unit rule used by the program

- If the maximum time value is smaller than 1, the program assumes seconds and converts the display to microseconds.
- If the maximum time value is 1 or larger, the program assumes the values are already in microseconds.
- Do not provide time in milliseconds. Convert it to seconds or microseconds before opening the files.

## Filename order

Files are sorted alphabetically. Use zero-padded numbers so that the experimental sequence is preserved: wave_001, wave_002, ..., wave_010 rather than wave_1, wave_2, ..., wave_10.

## Before opening the folder

- Keep only waveform files in the folder used for the picking session.
- Confirm that time increases monotonically and that each file contains several data points.
- Make a backup of the folder if it already contains an older waveform_picks_summary.txt file; saving will overwrite it.

# 3 Interface and picking workflow

| Control | What it does | Practical advice |
|---|---|---|
| Select Folder | Loads all valid files in the selected folder. | Select the waveform folder, not the folder containing the Python script. |
| Filter Cutoff | Sets the low-pass cutoff in MHz. | Start at 1.0 MHz and change it only when the signal requires it. |
| Apply Filter | Reloads the current waveform using the selected cutoff. | Compare the waveform before and after changing the cutoff. |
| P-wave / S-wave | Chooses which arrival will be assigned by the next click. | P is shown in red; S is shown in green. |
| Toolbar | Provides zoom, pan, home, and back-view controls. | Turn off Zoom or Pan before placing a pick. |
| Next Waveform | Moves forward to the next file. | There is no Previous button; verify both picks first. |
| Save All Picks | Writes all current picks to the summary file. | Unpicked arrivals are saved as NaN. |

## Recommended picking sequence

**1. Select the folder.** The first waveform appears and the file counter shows its position in the sequence.
**2. Inspect the trace.** Confirm that the horizontal axis is time in microseconds and that the waveform looks physically plausible.
**3. Choose the cutoff.** Use the highest cutoff that reduces unwanted high-frequency noise without removing the onset.
**4. Zoom around the expected P arrival.** Use the toolbar, then click the active Zoom button again to leave zoom mode.
**5. Pick P.** Select P-wave and click the first consistent departure from the pre-arrival noise. A red dashed line appears.
**6. Pick S.** Select S-wave and click the S onset. A green dashed line appears.
**7. Check the values.** The interface reports both picks in microseconds. P should normally precede S for the same propagation path.
**8. Advance.** Select Next Waveform only after both picks are satisfactory.
**9. Save.** Select Save All Picks at the end of the folder, or periodically during a long session.

# 4 How to choose an arrival

Pick the onset of coherent energy, not the largest peak. The onset is the first repeatable departure from the noise level that is consistent with neighboring waveforms and with the expected P- or S-wave time window.

## Quality-control rules

- Use the same arrival criterion for every waveform in the series.
- Check continuity from one waveform to the next; an isolated jump often indicates a poor pick.
- Use filtering to reveal the onset, not to force the waveform to match an expected answer.
- Record the cutoff used for the analysis. Changing the cutoff can change the apparent onset.
- If no reliable arrival exists, leave it unpicked. The output will contain NaN.

## About the filter

The GUI uses a fourth-order Butterworth low-pass filter and applies it forward and backward. This zero-phase procedure avoids the phase shift produced by a one-direction filter. If the requested cutoff exceeds the Nyquist frequency, the program automatically lowers it to 80 percent of Nyquist.

# 5 Output and interpretation

Selecting Save All Picks creates waveform_picks_summary.txt inside the waveform folder. The file is tab-separated and can be opened in a text editor, Excel, Python, MATLAB, or another analysis program.

| Output column | Meaning | Unit |
|---|---|---|
| Filename | Name of the waveform file | - |
| P_arrival_us | Picked P-wave arrival time | µs |
| S_arrival_us | Picked S-wave arrival time | µs |

Example output:

```
Filename P_arrival_us S_arrival_us
wave_001 12.4631      23.8176
wave_002 12.5210      23.9042
wave_003 NaN    24.0117
```

- NaN means that an arrival was not picked for that waveform.
- Saving again replaces the existing summary file with the current picks.
- The summary file is excluded when the folder is loaded again, so it is not treated as a waveform.

## Correct the apparatus delay before calculating velocity

The picked times are measured arrival times. They can include delays from the trigger, electronics, cables, transducers, coupling layers, and end blocks. Subtract the independently measured calibration delay before using the rock travel time. P and S paths may require different calibration values.

$$t_{P,\,\mathrm{rock}} = t_{P,\,\mathrm{pick}} - t_{P,\,0} \qquad t_{S,\,\mathrm{rock}} = t_{S,\,\mathrm{pick}} - t_{S,\,0}$$

After correction, calculate velocity using the propagation length L and travel time in consistent SI units (metres and seconds):

$$V_P = \frac{L}{t_{P,\,\mathrm{rock}}} \qquad V_S = \frac{L}{t_{S,\,\mathrm{rock}}}$$

# 6 Troubleshooting

| Problem | Likely cause | Action |
|---|---|---|
| ModuleNotFoundError | The environment is inactive or a package is missing. | Run conda activate waveform_picker, then repeat the installation command. |
| GUI does not open | Tk is unavailable in the environment. | Run conda install -c conda-forge tk, then relaunch. |
| No valid waveform files | The selected folder is empty or the files are inside subfolders. | Choose the folder that directly contains the waveform files. |
| Could not parse file | The first two columns are not consistently numeric or the delimiter is unusual. | Export a clean two-column text, CSV, or tab-separated file. |
| Time axis is incorrect | The source time is in milliseconds or mixed units. | Convert every file to seconds or microseconds before loading. |
| Click does not place a pick | Zoom or Pan mode is still active. | Click the active toolbar button again, then click the waveform. |
| Waveform is too noisy | The cutoff is too high or the raw signal has low signal-to-noise ratio. | Lower the cutoff gradually and compare neighboring traces. |
| Onset disappears | The cutoff is too low. | Increase the cutoff until the early waveform structure is preserved. |
| Files appear out of order | Alphabetical sorting differs from numeric sorting. | Rename files with zero-padded numbers such as 001, 002, 010. |
| Earlier pick needs correction | The GUI has no Previous button. | Restart the picking session and redo the sequence before saving the final file. |

# Final checklist

- ☐ The waveform_picker environment is active.
- ☐ Every waveform uses two numeric columns: amplitude first, time second.
- ☐ Time units are consistent and are either seconds or microseconds.
- ☐ The filter cutoff has been recorded and applied consistently.
- ☐ P and S picks follow the same onset criterion throughout the series.
- ☐ The summary file has been opened and checked for missing or implausible values.
- ☐ The appropriate apparatus delay has been subtracted before calculating velocity.